

Namma-Vastra

Weaver-to-Market Trend Guide

Project Report

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Technology Used: Android Studio, Kotlin, XML, Firebase

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I would also like to express my deep gratitude to the faculty members and department staff of my institution for encouraging me to participate in internship-based learning activities. Their academic support and technical knowledge helped me understand the concepts of Android application development, user interface design, database integration, and software

Abstract

Namma-Vastra is an Android application developed for supporting traditional Ilkal and Molakalmuru saree weavers.

The application helps weavers identify trending city fashion colors, upload saree products directly to an online gallery, calculate fair retail pricing, and preserve cultural weaving stories.

The project reduces dependence on wholesalers and supports self-employment opportunities through digital technology.

Namma-Vastra is an Android application developed to support traditional Ilkal and Molakalmuru saree weavers by helping them connect directly with modern fashion markets. Many weavers face difficulties because they mainly depend on wholesalers, who take a large share of the profit. In addition, most weavers are unaware of the latest fashion trends, modern color combinations, and customer preferences in urban areas. This results in low sales and reduced income for the weaving community.

The main objective of the Namma-Vastra application is to provide a digital platform where weavers can learn about trending saree colors, modern designs, and popular fashion styles. The application includes a Trend Board section that displays current fashion inspirations and color palettes suitable for handloom sarees. It also provides a Loom Gallery feature where weavers can upload images of their newly woven sarees and showcase them to boutiques and customers.

Another important feature of the application is the Material Costing module, which helps weavers calculate a fair retail price based on the cost of silk, cotton, and other materials used in saree production. The application also contains a Weaver Story section that documents the history, culture, and traditional weaving techniques of Ilkal and Molakalmuru sarees.

The project was developed using Android Studio with Kotlin programming language and XML for user interface design. Firebase services are used for authentication, image storage, and database management. The application is designed with a simple and user-friendly interface so that weavers can use it easily.

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Chapter 1

Introduction

India is well known for its rich cultural heritage, traditional arts, and handloom weaving industry. Among the many traditional weaving styles in Karnataka, Ilkal and Molakalmuru sarees are highly famous for their unique designs, craftsmanship, and cultural importance. These sarees are completely handmade by skilled weavers who spend several days creating each piece with dedication and hard work. Even though the sarees are valuable and admired by many people, the weavers often face financial difficulties and do not receive fair profits for their work.

One of the major problems faced by traditional weavers is their dependence on wholesalers and middlemen. Most weavers sell their sarees to wholesalers at low prices, and the wholesalers later sell them in urban markets at much higher prices. As a result, the weavers receive only a small portion of the actual profit. Another major challenge is the lack of awareness about modern fashion trends and customer preferences. Urban customers today prefer modern colors such as pastel shades, grey combinations, minimal patterns, and trendy designs. However, many traditional weavers continue producing old color combinations because they do not have access to current market information.

To solve these problems, the project “Namma-Vastra – Weaver-to-Market Trend Guide” was developed as an Android application. The main purpose of this application is to help weavers connect directly with modern fashion markets and improve their business opportunities. The application provides information about trending saree colors, modern design inspirations, and urban fashion preferences through a feature called Trend Board. This helps weavers understand what customers currently like and demand in the market.

Chapter 2

Problem Statement

Traditional weavers face several challenges:

- Dependence on wholesalers
- Lack of awareness about modern fashion trends
- Low retail profit
- Limited online visibility
- Difficulty connecting with urban boutiques

Chapter 3

Objectives

- Provide modern saree trend insights
- Allow direct saree showcasing
- Suggest fair pricing
- Preserve weaving culture
- Improve self-employment opportunities

Chapter 4

Technology Stack

4.1 Frontend

- XML Layouts
- Material UI
- RecyclerView
- CardView

4.2 Backend

- Firebase Authentication
- Firebase Firestore
- Firebase Storage

4.3 Programming Language

- Kotlin
- XML

Chapter 5

Modules of the Application

5.1 Trend Board

Displays trending colors and modern saree inspirations.

5.2 Loom Gallery

Allows weavers to upload saree images publicly.

5.3 Material Costing

Calculates fair retail price using silk and cotton prices.

5.4 Weaver Story

Documents the history and cultural importance of weaving traditions.

Chapter 6

System Requirements

6.1 Hardware Requirements

- Intel i5 Processor or higher
- Minimum 8GB RAM
- Android Smartphone

- Internet Connectivity

6.2 Software Requirements

- Android Studio
- Kotlin SDK
- Firebase
- JDK 17

Chapter 7

Project Structure

```
app/      java/

          MainActivity.kt
          TrendBoardActivity.kt
          LoomGalleryActivity.kt
          MaterialCostingActivity.kt
          WeaverStoryActivity.kt

          res/

          layout/ drawable/ values/
```

Chapter 8

MainActivity Kotlin Code

```
package com.example.nammavastra

import android.content.Intent
import android.os.Bundle
import android.widget.LinearLayout
import androidx.appcompat.app.AppCompatActivity

class MainActivity : AppCompatActivity() {

    override fun onCreate(savedInstanceState: Bundle?) {

        super.onCreate(savedInstanceState)

        setContentView(R.layout.activity_main)

        val trendBoard = findViewById<LinearLayout>(R.id.btnTrend)

        val loomGallery = findViewById<LinearLayout>(R.id.btnGallery)

        trendBoard.setOnClickListener {

            startActivity(
                Intent( this,
                    TrendBoardActivity::class.java
                )
            )
        }
    }
}
```

```
    )  
}  
  
loomGallery.setOnClickListener {  
  
    startActivity(  
        Intent( this,  
                LoomGalleryActivity::class.java )  
    ) }  
  
}  
}
```

Chapter 9

MainActivity XML Code

```
<?xml version="1.0" encoding="utf-8"?>

<LinearLayout xmlns:android=
    "http://schemas.android.com/apk/res/android"

    android:layout_width="match_parent"

    android:layout_height="match_parent"

    android:orientation="vertical"

    android:padding="16dp">

    <TextView

        android:layout_width="wrap_content"

        android:layout_height="wrap_content"

        android:text="Namma-Vastra"

        android:textSize="28sp"

        android:textStyle="bold"/>

</LinearLayout>
```

Chapter 10

Firestore Integration

```
implementation 'com.google.firebase:firebase-auth:22.3.0'
```

```
implementation 'com.google.firebase:firebase-firestore:24.10.0'
```

```
implementation 'com.google.firebase:firebase-storage:20.3.0'
```

6

Chapter 11

Advantages

- **Easy to use**

The application is designed with a simple and user-friendly interface so that weavers can easily understand and use all the features without technical difficulty.

- **Modern UI design**

The app uses attractive layouts, modern color combinations, and organized screens to provide a better user experience and professional appearance

- **Supports direct marketing**

Weavers can directly showcase their sarees to customers and boutiques without depending completely on wholesalers or middlemen.

- **Helps preserve culture**

The application promotes and documents the traditional weaving history, patterns, and cultural importance of Ilkal and Molakalmuru sarees

- **Supports self-employment**

By improving market visibility and sales opportunities, the app helps weavers increase their income and become financially independent

Chapter 12

Future Enhancements

- AI Trend Prediction
- Voice Support in Kannada
- Online Payment Gateway
- WhatsApp Integration
- Direct Buyer Chat

Chapter 13

Testing

13.1 Unit Testing

Unit testing is the process of testing individual modules or components of an application separately to ensure that each feature works correctly. In the Namma-Vastra application, unit testing was performed for important modules such as Trend Board, Loom Gallery, Material Costing, and Weaver Story sections.

The Trend Board module was tested to verify whether trending colors and fashion updates were displayed properly on the screen. The Loom Gallery module was tested to check image upload functionality and gallery display performance. The Material Costing module was tested by entering different silk and cotton prices to confirm that the application calculated the fair retail price accurately. The Weaver Story section was tested to ensure that historical content and cultural information were displayed correctly without errors.

Through unit testing, coding errors and functional issues were identified and corrected during the development process. This improved the reliability, stability, and overall performance of the application.

13.2 UI Testing

UI Testing (User Interface Testing) is performed to verify whether the application interface is user-friendly, visually clear, and functioning properly on different Android devices and screen sizes. In the Namma-Vastra application, UI testing was conducted to ensure that all screens, buttons,

layouts, images, and navigation features worked smoothly without any design issues.

The home screen, Trend Board, Loom Gallery, Material Costing, and Weaver Story modules were tested individually to check proper alignment, responsiveness, text visibility, and button functionality. Navigation between activities was also tested to ensure that users could easily move from one section to another without errors or delays.

Special attention was given to the simplicity of the interface so that traditional weavers with basic smartphone knowledge could use the application comfortably. UI testing helped improve the overall user experience, application appearance, and accessibility of the project.

Chapter 14

Screenshots

14.1 Home Screen

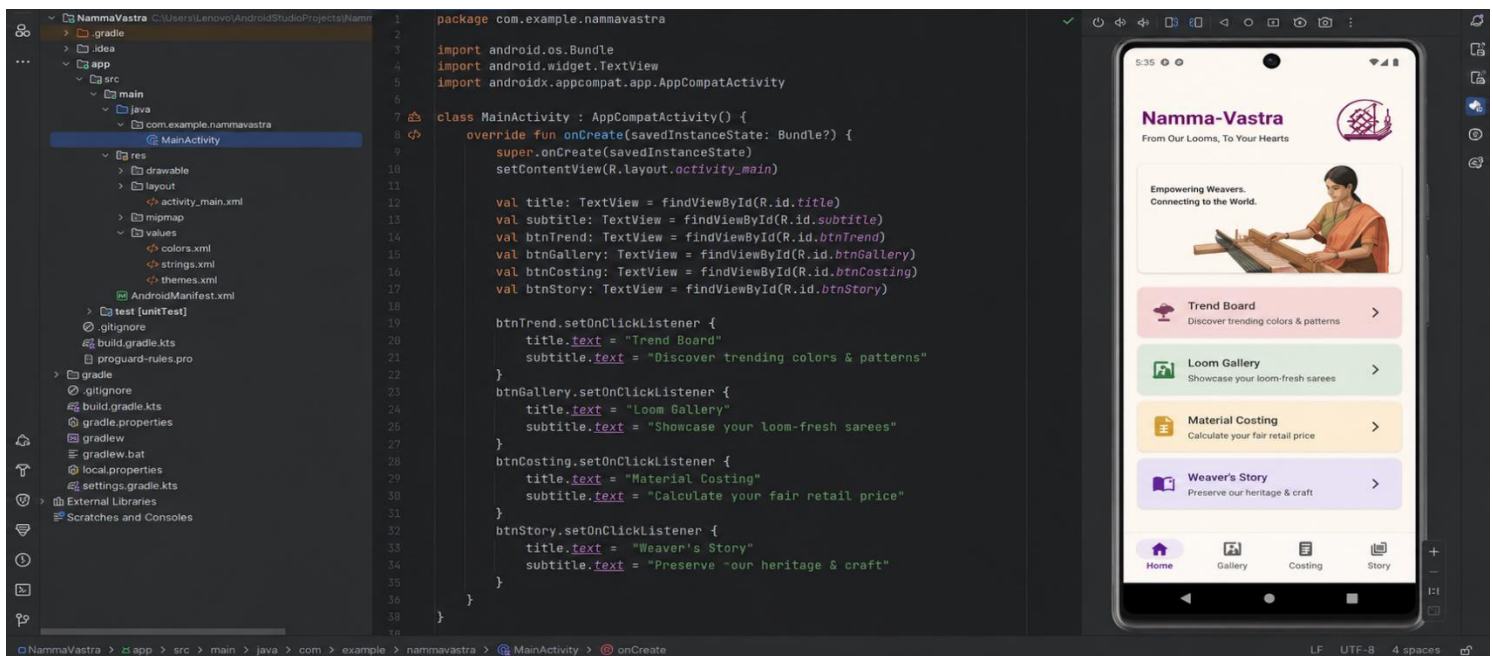
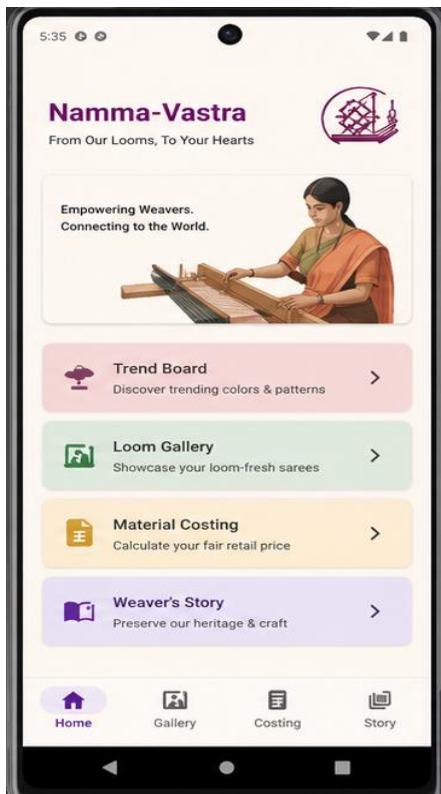


Figure 14.1: Home Screen



14.2 Trend Board

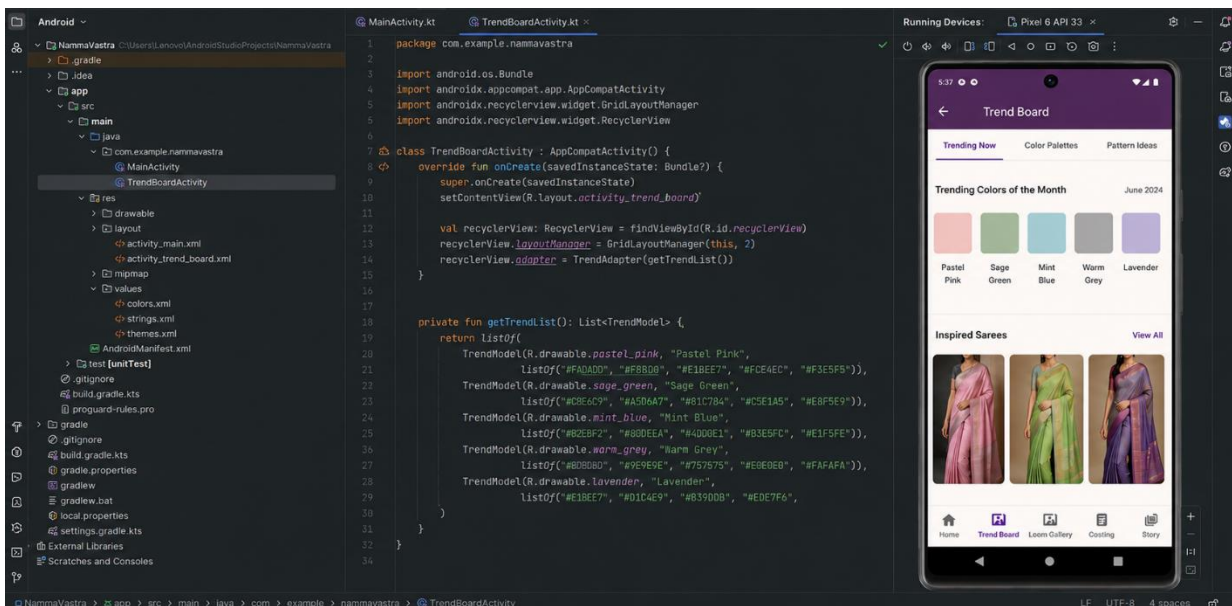
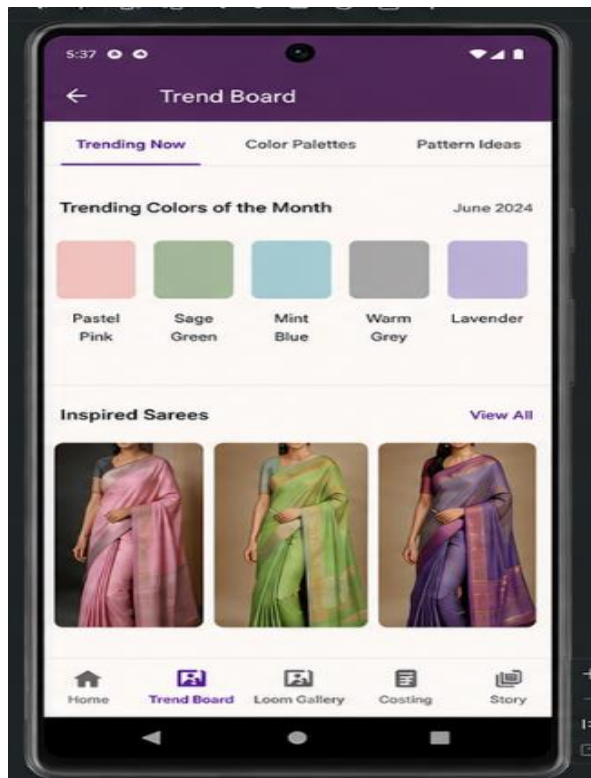
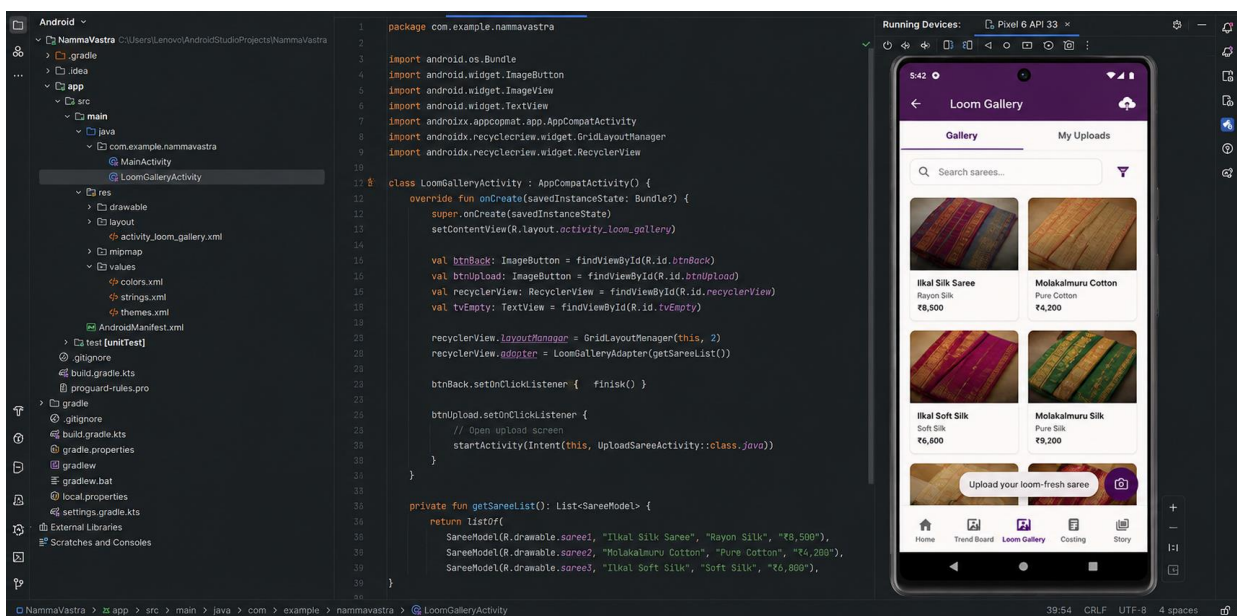


Figure 14.2: Trend Board



14.3 Loom Gallery



Chapter 15

Conclusion

Namma-Vastra successfully connects traditional weavers with modern fashion trends and urban buyers.

The application supports self-employment, improves visibility for handcrafted sarees, and preserves cultural heritage through technology.

Chapter 16

References

- [Android Developer Documentation](#)
- [Firebase Documentation](#)
- [Kotlin Programming Guide](#)
- [Material Design Guidelines](#)